

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.N o	Requirement Category	Requirement Description	Priority
1	Authentication	Current version does not require user login. Future versions may support secure authentication for personalized learning history.	Low
2	Authorization Levels	The current system provides a single user role with access to all available features. Future versions may include Student and Teacher roles.	Low
3	External Interfaces	The system integrates with Hugging Face Transformers for the BERT and BiLSTM models, and Google Gemini AI for generating personalized study guidance.	High
4	Transaction Processing	The system processes user inputs (subject and learning problem text), generates emotion predictions and AI guidance, and stores session history.	High
5	Reporting	The system generates a downloadable AI report containing detected emotions, confidence scores, and the personalized study plan.	Medium
6	Business Rules	Users must select a subject and enter a learning problem. The system detects primary and mixed emotions before generating guidance.	High
7	Compliance to Laws or Regulations	The application does not collect sensitive personal information. It follows general data privacy principles while storing session history in CSV format.	Medium
8	Other	The system stores emotion detection results and AI guidance history for future reference, analytics, and improvement.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate emotion predictions and AI guidance quickly.	Average response time < 3 seconds
2	Scalability	The system should support additional students and future cloud deployment.	Supports at least 100 concurrent users in future deployment
3	Security & Data Privacy	Student inputs, emotion data, and API interactions should be securely handled.	No sensitive personal data stored; secure API communication
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing
5	Usability & Accessibility	The interface should be intuitive and easy for students to navigate.	Students can get personalized guidance within 3 interaction steps
6	Maintainability & Portability	The code should be modular, reusable, and easy to deploy across different environments.	Modular architecture with reusable model and API components